PRODUCT REQUIREMENTS DOCUMENT

Human Resource Management System

Production-grade web platform | Business, functional, technical and operational requirements

Document Control	Details
Version	1.2 — Multi-tenant architecture baseline, configurable business rules, policy versioning, and scope alignment added
Status	Approved Architecture Baseline & Functional Specification
Prepared by	IT Department / Product Owner / Architecture Review Board
Date	15 October 2026
Primary audience	Management, HR, Finance, IT, Engineering, QA and Operations
Product direction	Multi-Tenant Enterprise HRMS SaaS Platform — shared application with strict tenant isolation, configurable policy engine, and modular core
Proposed stack	React + Vite; Spring Boot / Java 21; PostgreSQL; REST APIs; JWT-based authentication (subject to architecture approval)
Deployment decision	AWS / Airtel Cloud / other approved hosting — pending technical and commercial evaluation

Confidentiality: This document contains internal product planning information. It is intended for authorized stakeholders and should be reviewed before external distribution.

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1. Executive Summary

The Human Resource Management System (HRMS) is proposed as a production-grade web application to centralize employee information and standardize core HR workflows across employee lifecycle management, attendance, leave, payroll preparation, employee self-service, reporting and administration.

The product is intended to replace fragmented records and inconsistent workflows with a controlled, auditable application that provides role-appropriate access, consistent data relationships and clear workflow ownership. It must be designed for real operational use, including security, data integrity, recovery, monitoring, testing and maintainability—not merely for visual demonstration.

The initial release should prioritize the core HR foundation and high-frequency workflows. Payroll features must be scoped carefully: salary structures and payroll processing can be implemented, but statutory calculations and filings require validated company rules, current legal requirements and Finance sign-off before production use.

1.1 Product Summary

Dimension	Product direction
Product	Multi-tenant cloud HRMS SaaS platform for employee and HR operations across multiple customer organizations
Primary users	Employees, HR administrators, managers, payroll/Finance users, system administrators
Frontend	React + Vite; responsive, accessible enterprise UI
Backend	Spring Boot on Java 21; REST APIs; layered modular architecture
Database	PostgreSQL with tenant scoping, versioned migrations, relational constraints, composite indexes and transactional operations

Dimension	Product direction
Authentication	Secure login and session/token lifecycle; JWT is a proposed option subject to security design
Authorization	Server-enforced tenant-isolated role-based and record-level access control with configurable permissions
Deployment	Cloud-hosted production environment; provider selected after cost, region, security and operational review
Product delivery	Phased releases with acceptance criteria, QA gates and documented approvals

1.2 Executive Outcomes

- One authoritative employee record with defined ownership and lifecycle status.
- Traceable attendance correction and leave approval workflows.
- Consistent salary structure and payroll-period processing with reconciled totals.
- Role-appropriate employee self-service and manager views.
- Operational dashboards and reports based on validated source data.
- Security, auditability, backup, recovery and deployment practices appropriate for production.
- An extensible foundation for future integrations without committing to unapproved scope.

2. Product Vision, Goals and Success Metrics

2.1 Vision

Deliver a dependable, easy-to-use HR platform that gives employees access to relevant HR services, gives HR a controlled operational workspace, and gives management reliable visibility into workforce processes.

2.2 Product Goals

- Centralize employee master data and reduce avoidable duplicate records.
- Standardize request submission, review, approval, rejection and history.
- Make policy-driven rules explicit, testable and configurable where appropriate.
- Provide reliable reports with consistent definitions and traceable filters.
- Protect confidential employee and payroll information through least-privilege access.
- Support maintainable, observable releases and recoverable production operations.

2.3 Proposed Success Metrics

Metric	Definition	Target / baseline approach
Employee record completeness	Percentage of active employee records meeting approved mandatory-field rules	Set target after HR validates mandatory fields
Workflow completion	Percentage of submitted requests reaching a valid terminal state	Establish baseline during pilot
Approval turnaround	Elapsed time from request submission to final decision	Set SLA by request type and approver
Data reconciliation	Difference between source records and dashboard/report totals	Zero unexplained differences in agreed reports
Payroll reconciliation	Difference between component totals and payroll summary totals	Zero arithmetic discrepancies in tested scenarios
Availability	Production service availability measured over an agreed period	SLO to be approved after hosting architecture
Recovery readiness	Successful completion of scheduled backup and restore tests	Restore test required before production sign-off
Security defects	Open critical/high security findings at release gate	No unresolved critical/high findings without formal risk acceptance
User adoption	Active usage by intended users during pilot	Baseline and target agreed with HR/management

Targets must not be invented to make the document look stronger. Product and service-level targets should be approved once operational context, baseline data and hosting design are known.

3. Business Context and Problem Statement

3.1 Problem Statement

HR operations can become difficult to coordinate when employee details, attendance information, leave requests, salary inputs and approvals are maintained in disconnected tools or manual records. This can cause repeated data entry, inconsistent status visibility, unclear ownership and avoidable reconciliation work.

3.2 Business Needs

- Maintain a reliable employee master record with clear lifecycle history.
- Provide transparent approval ownership and request status.
- Keep historical records available when employees or organizational structures change.

- Make payroll inputs and calculations reviewable before a run is finalized.
- Produce consistent reports from the same underlying data and business definitions.
- Protect personal, employment and compensation information.
- Support controlled changes to policies, roles and application configuration.

3.3 Expected Benefits and Measurement

Expected benefits include improved consistency, reduced manual follow-up, better traceability and more reliable management reporting. These are hypotheses to validate during pilot usage; the project should measure actual effort, turnaround time, error rates and adoption rather than claim unverified savings.

4. Users, Personas and Role Model

Persona	Primary needs	Typical permissions
Employee	Own profile, attendance, leave balance, requests, payslips and notifications	Self-service access to own records; submit and track requests
HR Administrator	Employee lifecycle, organization data, policies, attendance exceptions, leave administration and HR reports	Manage HR records and workflows within assigned scope
Department Manager	Team visibility and assigned approval queues	View permitted team records; approve/reject assigned requests
Payroll / Finance User	Salary structures, payroll inputs, payroll runs, reconciliations and payslips	Access compensation and payroll features based on approved scope
System Administrator	Users, roles, configuration, integration settings and operational support	Administrative controls; sensitive data access should be restricted by need
Auditor / Read-only Reviewer	Traceable records, approved reports and audit events	Read-only, time-bounded access if required
Management Viewer	Workforce KPIs and approved summary reports	Aggregated reporting; no default access to unnecessary personal details

4.1 Authorization Principles

- All access decisions must be enforced by the backend; hiding a button in the frontend is not authorization.
- Apply least privilege and deny access by default.
- Separate system administration from payroll and HR business access where practicable.
- Restrict employee self-service to the authenticated employee's own records.
- Scope manager access to assigned teams or reporting relationships.
- Record security-sensitive administrative actions in an audit trail.

- Use a formal process to grant, review, change and revoke access.

4.2 Proposed Roles

Initial role names are proposed and must be confirmed with management and HR: SUPER_ADMIN, HR_ADMIN, MANAGER, PAYROLL_ADMIN, EMPLOYEE, AUDITOR_READONLY and MANAGEMENT_VIEWER. Permissions should be granular enough to avoid giving every administrator unrestricted access to all confidential records.

5. Product Scope and Release Boundaries

5.1 Proposed Release 1 — Core Production HRMS

Module	Release 1 scope	Priority
Identity and access	Login/logout, password or approved identity flow, role assignment, access enforcement, account lifecycle	P0
Tenant architecture & Organization setup	Tenant boundary and context enforcement, tenant-specific settings, multi-branch/locations, departments, designations, reporting relationships, holidays, and tenant-level feature entitlements.	P0
Employee lifecycle	Employee profiles, employment details, onboarding checklist, status changes, document metadata	P0
Attendance	Attendance records/import-ready structure, shifts, attendance exceptions, regularization requests	P0
Leave	Leave types, policy rules, balances, requests, approval workflow, history	P0
Payroll foundation	Salary components/structures, payroll periods, inputs, calculation engine boundary, draft/review/finalize states	P0
Payslips	Controlled payslip generation and employee access after approved payroll finalization	P0
Reports	Employee, attendance, leave and payroll summary reports with role-based access	P0
Notifications	In-app notifications for supported workflow events	P1
Audit and administration	Audit events, settings, role administration and operational configuration	P0
Deployment operations	Environment separation, CI/CD, backups, monitoring, logs and restore procedures	P0
Policy & Workflow Engine	Versioned policy definitions (attendance late/grace/regularization, leave accrual/carryover), policy assignments by branch/dept/group, and configurable multi-step approval workflows with delegation & escalation.	P0

5.2 Potential Later Releases

- Biometric/attendance-device integration after device and vendor selection.
- Email/SMS/WhatsApp notifications after provider, cost and consent review.
- Accounting/ERP integration after interface and ownership are agreed.
- Advanced analytics, workforce planning and configurable dashboards.
- Mobile application or progressive web application enhancements.
- Advanced recruitment, performance management, learning, expenses or asset management only if explicitly approved as product scope.

5.3 Out of Scope Unless Approved

- Unapproved third-party integrations or paid provider commitments.
- Automatic legal/tax filing without validated rules and authorized integration.
- Bank payment execution or direct disbursement in the initial release unless explicitly required and approved.
- Features unrelated to HRMS core objectives that materially increase release risk.
- Treating prototype-only shortcuts as acceptable production security controls.

6. Information Architecture and Navigation

Navigation group	Primary pages
Home	Role-specific dashboard, assigned tasks, alerts and recent activity
People	Employee directory, employee profile, onboarding, departments, designations, branches, reporting structure, documents
Time & Attendance	Attendance register, daily detail, shift roster, holidays, regularization queue, imports and exceptions
Leave	Leave overview, policy configuration, balances, requests, approval queue and history
Payroll	Salary components, salary structures, payroll periods, payroll runs, exceptions, payslips and payroll reports
Reports	Employee, attendance, leave, payroll and audit reports permitted for the user's role
Administration	Users, roles, permissions, company settings, integrations, notification settings and audit logs
My Workspace	My profile, my attendance, my leave, my requests, my payslips and personal notifications

The UI should use a consistent design system and original visual identity. The previously discussed palette—purple #AA00EA, deep purple #310052, cyan #A3FAFF and light neutral backgrounds—may be retained as a design direction, with accessible contrast and restrained accent usage.

7. Functional Requirements by Module

Priority: P0 = mandatory for the agreed first production release; P1 = important but may follow the core release; P2 = future candidate. Each requirement must be linked to implementation tickets and test cases.

7.1 Identity, Authentication and Access

ID	Requirement	Priority	Acceptance summary
FR-IAM-001	Login and logout	P0	Users can authenticate through the approved identity mechanism; logout invalidates the applicable session/token state.
FR-IAM-002	Account lifecycle	P0	Authorized administrators can activate, deactivate and review user accounts; deactivated accounts cannot continue to access protected APIs.
FR-IAM-003	Role and permission administration	P0	Authorized administrators assign approved roles and granular permissions; changes take effect according to documented token/session rules.
FR-IAM-004	Server-side authorization	P0	Every protected endpoint validates authentication, role and record scope; direct API calls cannot bypass UI restrictions.
FR-IAM-005	Credential recovery and protection	P0	If local credentials are used, implement secure password hashing, recovery workflow, rate limiting and safe error messages; exact policy requires approval.
FR-IAM-006	Session and token lifecycle	P0	Access token lifetime, refresh/re-authentication, revocation strategy and storage model are documented and security-reviewed.

7.2 Organization and Employee Lifecycle

ID	Requirement	Priority	Acceptance summary
FR-ORG-001	Company and branch setup	P0	Maintain approved company/branch/location records and effective dates where needed.
FR-ORG-002	Departments and designations	P0	Create, edit, deactivate and assign organization entities; prevent invalid references.
FR-EMP-001	Employee directory	P0	Search, filter, sort and paginate employee records according to permission scope.

ID	Requirement	Priority	Acceptance summary
FR-EMP-002	Employee profile	P0	Maintain identity, contact, employment, reporting, department, designation and lifecycle details with field-level validation.
FR-EMP-003	Unique employee identifier	P0	Employee identifier is unique within the defined organization scope; duplicate attempts return a clear validation error.
FR-EMP-004	Onboarding checklist	P1	Track required onboarding steps, owners, due dates and completion state.
FR-EMP-005	Employee status and separation	P0	Support active, inactive, separated and other approved statuses; preserve historical transactions.
FR-EMP-006	Employee documents	P1	Store document metadata and use secure object/file storage for actual files; access and retention rules are enforced.
FR-EMP-007	Employee import	P1	Provide validated bulk import with preview, row-level errors, duplicate detection and an import result report.

7.3 Attendance and Shift Management

ID	Requirement	Priority	Acceptance summary
FR-ATT-001	Attendance register	P0	View attendance by date, employee, department and status within the user's scope.
FR-ATT-002	Attendance source and provenance	P0	Each attendance record identifies whether it originated from manual entry, import, integration or approved correction.
FR-ATT-003	Shift configuration	P0	Configure shift start/end times, break rules, overnight handling and effective dates according to approved policy.
FR-ATT-004	Attendance calculation	P0	Calculate worked duration and status using approved rules; handle overnight shifts, missing punches and invalid sequences.
FR-ATT-005	Regularization workflow	P0	Employee submits correction request; reviewer decides; original value, proposed value, reason, decision and timestamps remain traceable.
FR-ATT-006	Bulk attendance import	P1	Import supported CSV/XLSX format after template approval; validate rows and provide retryable error reporting.
FR-ATT-007	Attendance exceptions	P0	Flag missing punches, duplicate entries, unassigned shifts and other defined exceptions for authorized review.

ID	Requirement	Priority	Acceptance summary
FR-ATT-008	Holidays and weekly offs	P0	Maintain holiday calendars and weekly-off rules by approved scope and effective date.

7.4 Leave Management

ID	Requirement	Priority	Acceptance summary
FR-LEV-001	Leave type and policy setup	P0	Configure leave types, eligibility, entitlement, accrual and other approved policy parameters.
FR-LEV-002	Leave application	P0	Employee submits a date-range request with leave type and reason; server validates dates and eligibility.
FR-LEV-003	Balance calculation	P0	Available, accrued, used, pending and adjusted balances follow documented formulas and are consistent across screens.
FR-LEV-004	Approval routing	P0	Requests route to the configured manager/HR approver; missing or inactive approvers create a visible exception.
FR-LEV-005	Approval/rejection	P0	Only authorized approvers can act; rejection requires a reason; terminal requests cannot be decided again.
FR-LEV-006	Cancellation and amendment	P1	Define rules for pending and approved leave cancellation/amendment, including approval and balance reversal.
FR-LEV-007	Leave history and reports	P0	Employees and authorized reviewers can see status history and period summaries within permission scope.

7.5 Payroll and Payslips

ID	Requirement	Priority	Acceptance summary
FR-PAY-001	Payroll configuration	P0	Maintain approved pay frequency, payroll periods, components and rounding rules.
FR-PAY-002	Salary structures	P0	Maintain effective-dated earnings and deductions; changes do not silently alter finalized historical payroll.
FR-PAY-003	Payroll input collection	P0	Collect approved attendance, leave, variable pay, deductions and other inputs with provenance and review status.

ID	Requirement	Priority	Acceptance summary
FR-PAY-004	Calculation engine	P0	Calculate gross earnings, deductions and net pay using versioned, testable rules approved by Finance.
FR-PAY-005	Payroll run workflow	P0	Support draft, calculated, under review, approved/finalized and cancelled/reversed states as approved by Finance.
FR-PAY-006	Exception handling	P0	Flag missing salary data, negative/invalid values, missing inputs and reconciliation issues before finalization.
FR-PAY-007	Payroll reconciliation	P0	Component totals reconcile with employee-level and run-level totals; rounding rules are consistent.
FR-PAY-008	Payslip generation	P0	Generate a controlled payslip for finalized payroll only, with authorized access and correct employee/period mapping.
FR-PAY-009	Payroll correction	P1	Define controlled post-finalization adjustment or reversal with reason, approval and audit history.
FR-PAY-010	Statutory rules	P0	Applicable statutory calculations are implemented only after requirements, effective dates, tests and Finance/legal sign-off are approved.

7.6 Employee Self-Service

ID	Requirement	Priority	Acceptance summary
FR-ESS-001	Own profile	P0	Employee can view permitted personal/employment details and request changes where policy allows.
FR-ESS-002	Own attendance and leave	P0	Employee can view only their authorized attendance and leave information.
FR-ESS-003	Own payslips	P0	Employee can access only payslips associated with their authenticated identity and finalized payroll periods.
FR-ESS-004	Request tracking	P0	Employee can see request status, submission time, decision and permitted reviewer comments.

7.7 Reports, Dashboards and Notifications

ID	Requirement	Priority	Acceptance summary
FR-REP-001	Dashboard metrics	P0	Metrics are calculated from authoritative records and have documented definitions and filters.
FR-REP-002	Employee report	P0	Authorized users can filter employee data and export approved fields.
FR-REP-003	Attendance report	P0	Period totals reconcile with underlying attendance records and approved status rules.

ID	Requirement	Priority	Acceptance summary
FR-REP-004	Leave report	P0	Report includes approved, pending, rejected and cancelled states as defined by filters.
FR-REP-005	Payroll report	P0	Compensation details are restricted to authorized roles and totals reconcile with payroll runs.
FR-REP-006	Export controls	P0	Exports respect the same row/field permissions as the UI and are logged where sensitive.
FR-NOT-001	In-app notifications	P1	Supported workflow events create notifications for the correct recipient and link to an authorized destination.
FR-NOT-002	Notification preferences	P2	Users can configure supported preferences where approved.

7.8 Audit, Administration and Support

ID	Requirement	Priority	Acceptance summary
FR-AUD-001	Business audit trail	P0	Record actor, action, target, timestamp, outcome and relevant before/after values for defined sensitive events.
FR-AUD-002	Audit access	P0	Audit events are append-oriented, access-controlled and protected from ordinary user modification.
FR-ADM-001	System configuration	P0	Authorized administrators can manage approved settings with validation and change history.
FR-ADM-002	Health and readiness endpoints	P0	Provide appropriately protected health/readiness checks for deployment and monitoring.
FR-ADM-003	Operational error handling	P0	Return consistent API errors with correlation IDs; do not expose secrets, stack traces or sensitive data.
FR-ADM-004	Support diagnostics	P1	Authorized support users can find request IDs and operational events without unrestricted access to confidential business records.

8. End-to-End Business Workflows

8.1 Employee Onboarding

HR creates an employee record and enters mandatory identity and employment fields.

The server validates required fields, uniqueness, organization references and permitted status transitions.

HR assigns department, designation, manager, work location and effective date.

Required onboarding checklist items and document metadata are recorded if enabled.

An authorized reviewer verifies the record where the approved process requires review.

The employee account is provisioned separately from the employee master record, using the approved identity process.

Audit history records creation and sensitive changes; reports reflect the new employee.

8.2 Attendance Regularization

Employee selects an eligible attendance date and submits the proposed correction and reason.

Server validates the employee's access, request eligibility, duplicate pending requests and policy constraints.

Request is routed to the assigned approver; the original attendance value remains unchanged while pending.

Approver reviews original data, proposed change and reason; approval or rejection is recorded with timestamp and identity.

On approval, the system applies the correction transactionally and links the changed record to the request.

Related summaries are recalculated or queried from the updated source; the employee can see the outcome.

8.3 Leave Request and Approval

Employee selects leave type and date range; system shows applicable balance and policy notes.

Server validates dates, eligibility, overlap, balance rules and any configured restrictions.

Request is stored as pending and routed to the correct approver.

Approver reviews team coverage and request details, then approves or rejects with a reason where required.

Approval and balance updates occur atomically to prevent inconsistent balances or duplicate decisions.

Cancellation, amendment, reversal and concurrent requests follow the approved leave policy.

Employee, HR and manager views display consistent status and history.

8.4 Payroll Processing

Payroll administrator opens a payroll period and verifies period boundaries and eligible employees.

System gathers effective-dated salary structures and approved inputs from attendance, leave and other configured sources.

Calculation engine runs the approved, versioned rules and stores a calculation snapshot for review.

Validation checks identify missing inputs, invalid components, outliers and reconciliation discrepancies.

Authorized payroll reviewer examines exceptions and approves the run according to the approval matrix.

Finalization locks the payroll snapshot from silent edits; corrections use a controlled adjustment or reversal process.

Payslips become available only after finalization and are accessible only to authorized users.

Audit records capture the run, calculation version, reviewer decisions and finalization event.

8.5 Employee Separation

Authorized HR user records separation type, effective date and approved checklist.

System checks outstanding workflow items and records required handover tasks.

Access revocation is coordinated with the identity/account lifecycle process.

Historical attendance, leave, payroll and audit records remain retained according to approved retention rules.

Final settlement is not assumed to be automated unless separately specified and validated.

9. Business Rules and Validation Catalogue

Area	Rule ID	Rule / decision required
Employee	BR-EMP-01	Employee ID uniqueness scope (company-wide or tenant-wide) must be approved.
Employee	BR-EMP-02	Employment dates and status transitions must be validated; historical transactions must be preserved.
Attendance	BR-ATT-01	Define punch pairing, missing punches, overnight shifts, breaks, grace periods, late/early rules and rounding.
Attendance	BR-ATT-02	Define attendance import format, duplicate detection, timezone and correction precedence.
Leave	BR-LEV-01	Define accrual, entitlement, carry-forward, expiry, negative balance, half-day and holiday/weekend treatment.
Leave	BR-LEV-02	Define overlapping requests, cancellation rules, backdated requests and approval delegation.
Payroll	BR-PAY-01	Define pay period, joining/leaving proration, unpaid leave treatment and variable-pay cut-off.

Area	Rule ID	Rule / decision required
Payroll	BR-PAY-02	Define component formulas, rounding, effective dating and treatment of retroactive changes.
Payroll	BR-PAY-03	Identify applicable statutory components, thresholds, exemptions and effective dates; validate with Finance/legal expertise.
Security	BR-SEC-01	Define password/session policy or approved SSO method, MFA requirements and account recovery.
Approval	BR-WF-01	Define approval chain, escalation, delegation, approver absence and segregation of duties.
Retention	BR-DATA-01	Define retention and deletion rules by data category, including legal holds if applicable.

No HR or payroll rule should be hard-coded from assumption when it materially affects salary, leave entitlement, attendance status or compliance. Rules must be approved, versioned where appropriate, and covered by test cases.

10. Data Model and Entity Relationships

10.1 Proposed Core Entities

Entity	Purpose / key relationships
Tenant / Organization	Company boundary, settings, timezone, policy defaults; parent scope for organization data if multi-tenancy is approved.
Branch / Location	Work location, timezone and location-specific calendars.
Department	Organizational unit; references organization and may have a parent department.
Designation	Position/title catalogue with active state.
Employee	Employee master, employment status, reporting manager, department, designation and location.
User Account	Authentication identity mapped to employee where applicable; distinct from employee master.
Role / Permission / UserRole	Authorization model and role assignment; server-enforced.
Shift / Shift Assignment	Shift definitions and effective-dated assignment to employees or groups.
Attendance Record	Employee/date attendance values, source, status and calculation metadata.
Attendance Import Batch	Import metadata, file hash/reference, validation results and row-level errors.
Regularization Request	Requested correction, original reference, reason, decision and approver.

Entity	Purpose / key relationships
Holiday Calendar / Holiday	Holiday rules and dates by organization/location.
Leave Type / Leave Policy	Leave category and versioned policy parameters.
Leave Balance Ledger	Accrual, usage, adjustment and reversal entries for explainable balance calculation.
Leave Request / Approval Step	Request dates, status, decision history and approval routing.
Salary Component	Earning/deduction definition and calculation metadata.
Salary Structure / Structure Component	Effective-dated component assignments for an employee or eligible group.
Payroll Period / Payroll Run	Period boundaries, state, calculation version, approval and finalization metadata.
Payroll Input / Payroll Line	Input provenance and employee-level earnings/deductions for a run.
Payslip	Finalized employee-period compensation snapshot and access metadata.
Notification	Recipient, event, status and destination reference.
Audit Event	Actor, event, target, timestamp, correlation ID and permitted change metadata.
File Metadata	Secure file reference, owner, category, checksum, scan status and retention metadata.
Organization Settings	Tenant-specific preferences: timezone, currency, fiscal year, working hours and policy defaults.
Feature Entitlement	Module licensing and capability toggles available to each customer tenant.
Policy Definition / Version	Canonical business rule definitions and effective-dated policy versions (attendance grace periods, late limits, leave accruals).
Policy Assignment	Hierarchical policy assignment to organization, legal entity, branch, department, shift, or employee group.
Approval Workflow / Step	Configurable multi-step approval sequences, approver roles (manager, HR, skip-level), escalation rules and delegation.

10.2 Data Design Principles

- Use stable primary keys and explicit foreign-key relationships.
- Use database constraints for uniqueness, referential integrity and required fields.
- Use transactions for approval decisions, balance updates and payroll state transitions.
- Use effective dates for policy, salary and assignment changes that affect historical calculations.
- Prefer ledger or transaction records for balances that require traceability.
- Use database migrations under source control; avoid manual, undocumented production schema changes.
- Define indexes based on real query patterns, including employee, date range, status and organization scope.
- Store timestamps consistently in UTC and render them in the configured user/organization timezone.

- Use decimal types and explicit rounding rules for monetary values; do not use floating-point types for money.
- Define retention, deletion, anonymization and export rules before production data is loaded.

10.3 Multi-Tenant Architecture and Configurable Business Rules

Per Architecture Review, the platform baseline is formally committed as a Multi-Tenant SaaS application. The system uses a shared application and database architecture with strict logical tenant isolation enforced at every layer: REST API request interceptors, Hibernate/JPA tenant scoping filters, background scheduling jobs, analytical reports, file storage namespaces, and audit logging. Adding a tenant_id column alone is necessary but not sufficient; tenant context must be immutable and resolved from the authenticated user token for every operation.

The Three-Tier Configuration Model

To prevent architectural conflation, the system strictly separates three distinct configuration dimensions:

1. Feature Entitlements (Platform Level): Defines which business modules and capabilities (e.g. Attendance, Leave, Payroll, Advanced Analytics) are licensed and enabled for a given customer tenant. This is managed by system administrators and determines API route and UI navigation availability.

2. Policy Configuration (Tenant & Organizational Level): Defines versioned rules for how business logic executes (e.g., attendance grace periods, regularization submission windows, leave accrual schedules, overtime multipliers). Policies are assigned hierarchically to legal entities, branches, departments, shifts, or employee groups.

3. Roles and Permissions (User Authorization Level): Defines what individual users can view or execute within their tenant. Permissions are granular and scoped by department, branch, and team reporting structure.

Policy Versioning and Historical Immutability Guarantee

A fundamental flaw in legacy HRMS implementations is allowing administrative edits to silently mutate historical calculations. GA-HRMS guarantees data integrity through versioned policy history:

• Effective Dating: Every policy definition (e.g., 'Standard Attendance Policy') maintains an immutable chain of policy versions with effective_from and effective_to dates.

• Historical Invariance: If a company increases its grace period from 10 minutes to 15 minutes starting on 1 November, attendance records prior to 1 November remain bound to the 10-minute policy version. Finalized attendance registers and approved payroll runs are cryptographically and transactionally locked against subsequent policy edits.

• Pending Request Evaluation: When a policy changes, the policy version specifies whether pending workflow requests follow the submission-time policy or are evaluated against the newly active version.

• Full Audit Provenance: Every policy creation, version update, and assignment change records the acting administrator, exact timestamp, IP address, and JSON diff of all modified parameters.

Multi-Tenant Architecture Acceptance Criteria

The following criteria are mandatory for the Phase 1 production baseline and must be verified by automated test suites:

• AC-MTC-01 (Tenant Isolation): Every API request, repository query, background batch job, file retrieval, and report export must enforce the active tenant boundary. Direct query execution without tenant scoping is prevented by automated database filters. Cross-tenant access attempts return HTTP 403 Forbidden and trigger high-priority security audit events.

• AC-MTC-02 (Default Policies & Tenant Overrides): The platform provides system-default policy templates while allowing each customer tenant to configure and override policy parameters independently without code changes or database redeployments.

• AC-MTC-03 (Effective-Dated Versioning): All attendance, leave, and payroll policies support versioning with future-dated and retroactive-safe boundaries. Edits create new versions; existing finalized historical transactions remain untouched.

• AC-MTC-04 (Role-Based and Record-Level Authorization): Authorization decisions are server-enforced, combining role permissions with record-level ownership checks (e.g., employees access only their own records; managers access only their assigned reporting lines).

• AC-MTC-05 (Configurable Approval Workflows): Workflow routing supports configurable multi-step approval sequences (e.g. Employee -> Manager -> Department Head -> HR Admin), approver delegation during absences, automated escalation after configurable SLA timeouts, and segregation of duties (a user cannot approve their own requests).

• AC-MTC-06 (Tenant Feature Entitlements): Module access is dynamically controlled via tenant feature entitlements. Disabled modules are completely hidden in the UI and reject API requests with standardized entitlement error codes.

• AC-MTC-07 (Backward-Compatible Shared Migrations): All database migrations must be forward- and backward-compatible to support zero-downtime shared releases across all hosted tenants.

• AC-MTC-08 (Automated Multi-Tenant Verification): CI/CD pipeline includes integration tests simulating multiple concurrent customer tenants executing differing attendance, leave, and payroll policies simultaneously, validating zero cross-tenant contamination.

11. API and Integration Requirements

11.1 API Standards

- Use versioned REST endpoints (for example, /api/v1) with consistent resource naming.
- Validate all input on the server; frontend validation is supplementary only.
- Return consistent status codes and a standard error body containing a safe message, stable error code and correlation ID.

- Use pagination, sorting and allow-listed filtering for list endpoints.
- Define request/response schemas and document APIs with OpenAPI/Swagger.
- Use idempotency or duplicate-request protections for operations where retries could create duplicate records.
- Apply transaction boundaries to multi-step business operations.
- Never expose passwords, secrets, internal stack traces or unnecessary sensitive fields in API responses or logs.
- Protect exports and file endpoints with the same authorization rules as normal record views.
- Apply rate limits and abuse protection to authentication and other sensitive endpoints.

11.2 Suggested Resource Catalogue

Resource	Example endpoint family	Typical operations
Authentication	/api/v1/auth/*	Login, logout, refresh/re-authentication, password recovery if supported
Users and roles	/api/v1/users, /api/v1/roles	Administer accounts, roles and permissions
Employees	/api/v1/employees	List, create, retrieve, update, change status
Organization	/api/v1/departments, /designations, /locations	Manage organization records
Attendance	/api/v1/attendance	List records, review details, import or calculate according to policy
Regularization	/api/v1/attendance-requests	Submit, review, approve, reject and track requests
Leave	/api/v1/leave-types, /leave-requests, /leave-balances	Configure policies, submit requests, review and retrieve balances
Payroll	/api/v1/payroll-periods, /payroll-runs	Prepare, calculate, review and finalize runs
Payslips	/api/v1/payslips	List and retrieve authorized finalized payslips
Reports	/api/v1/reports/*	Generate authorized report data and exports
Notifications	/api/v1/notifications	List, mark read and retrieve permitted notifications
Audit	/api/v1/audit-events	Restricted read-only search for authorized reviewers
Health	/actuator/health/* or approved equivalent	Operational readiness and health checks with exposure controls

11.3 Integration Boundaries

Potential integrations include attendance devices, email delivery, accounting systems, identity providers and secure document storage. Each integration requires a defined business owner, vendor/API selection,

authentication model, error/retry policy, reconciliation procedure, data mapping, cost review and acceptance tests. No external provider should be treated as committed until approved.

12. Security, Privacy and Compliance

12.1 Security Requirements

- Enforce TLS for all production traffic and secure transport between trusted services where applicable.
- Store passwords only using a modern adaptive password-hashing algorithm if local authentication is selected.
- Protect access and refresh tokens according to a reviewed token lifecycle and browser threat model.
- Use least-privilege roles and record-level authorization for employee, manager and payroll data.
- Validate inputs and encode outputs; use parameterized database access and safe file-handling controls.
- Protect against common web risks including injection, broken access control, CSRF where relevant, XSS, SSRF, insecure file upload and misconfiguration.
- Keep secrets in an approved secrets manager or secure deployment configuration, never in source control.
- Encrypt database, file and backup storage using provider-supported encryption and controlled key access.
- Log security events while minimizing personal data and avoiding credentials or full sensitive payloads.
- Apply dependency scanning, patching, secure code review and vulnerability management.
- Restrict production administrative access and require stronger authentication for privileged users where feasible.
- Define incident response, access review and credential rotation procedures.

12.2 Privacy and Data Governance

- Collect only data needed for approved HR purposes.
- Define purpose, access, retention, correction and deletion rules for each sensitive data category.
- Provide appropriate access to employees for their own permitted information.
- Restrict payroll exports and document downloads to authorized roles.
- Use synthetic or anonymized data in development and test environments.
- Record and review access to particularly sensitive operations.
- Assess applicable Indian data protection, employment, tax and record-retention obligations with qualified internal/external advisors before production launch.
- Define how data subject requests, incidents, vendor access and data transfers will be handled where applicable.

12.3 Audit Events

At minimum, audit employee creation/changes/status transitions, role and permission changes, attendance corrections, leave decisions, salary structure changes, payroll calculation/finalization/reversal, sensitive exports, administrative configuration changes and relevant authentication/security events. Audit records should include

actor, event type, target identifier, timestamp, outcome and correlation ID, with carefully controlled before/after values.

13. Non-Functional Requirements

Category	Requirement	Acceptance approach
Availability	Define production availability objective after hosting and support model are chosen.	Monitor uptime and incident duration; report against approved SLO.
Performance	Core pages and APIs should meet agreed response-time targets at expected concurrency and data volume.	Load-test representative APIs and list/report queries; approve numeric thresholds before release.
Scalability	Application should support growth through measured capacity planning and horizontal scaling where appropriate.	Test agreed peak workload and identify database/application bottlenecks.
Reliability	Failures should be handled predictably without corrupting business transactions.	Test retries, transaction rollback, duplicate requests and dependency failures.
Data integrity	Related records and financial calculations must remain consistent.	Constraint tests, transaction tests and reconciliation suites.
Security	No unresolved critical/high findings at production release without formal risk acceptance.	Threat modelling, SAST/dependency scans, access-control tests and penetration testing as appropriate.
Accessibility	Common workflows should support keyboard use, labels, readable contrast and clear errors.	Manual and automated accessibility checks against an agreed standard.
Compatibility	Support an agreed browser/device matrix.	Regression testing on approved browser versions and viewport sizes.
Maintainability	Use modular code, consistent standards, documentation and automated tests.	Code review, coverage of critical logic and maintainability checks.
Observability	Provide structured logs, metrics, traces where appropriate and alerting.	Operational dashboard and alert verification.
Recoverability	Backups and restores must be tested; RPO/RTO targets must be approved.	Documented restore drill and evidence before production sign-off.
Localization/time	Use configured locale, currency, timezone and date formats consistently.	Tests for date boundaries, DST/timezone handling where relevant, and monetary formatting.

14. Production Architecture and Technology

14.1 Proposed Logical Architecture

Layer	Proposed technology / responsibility
Web client	React + Vite; responsive UI, client-side routing, accessible components and API integration.
API / application	Spring Boot on Java 21; REST controllers, DTO validation, application services and centralized error handling.
Domain layer	Business rules for employee lifecycle, attendance, leave, approval workflows and payroll orchestration.
Persistence	PostgreSQL; relational schema, constraints, transactions, indexes and controlled migrations.
Data access	Spring Data JPA/Hibernate or another approved persistence approach; query patterns reviewed for performance.
Authentication	JWT or another approved mechanism; token storage, expiry, refresh and revocation designed explicitly.
File storage	Private object storage or secure managed file service for documents and generated payslips, subject to provider choice.
Notifications	In-app notification records initially; external email/SMS provider selected only after approval.
Operations	CI/CD, environment configuration, secret management, monitoring, logs, backups and recovery procedures.

14.2 Architectural Principles

- Start with a modular monolith unless scale, team structure or integration constraints justify distributed services; avoid premature microservices.
- Keep business logic out of controllers and UI components.
- Separate API contracts from persistence entities.
- Use database transactions for workflows that update multiple related records.
- Keep payroll calculations deterministic, versioned and independently testable.
- Use database migrations as the source of truth for schema evolution.
- Separate development, staging/UAT and production environments.
- Externalize configuration and secrets; never commit production credentials.
- Use asynchronous jobs only where needed and ensure retry/idempotency behaviour is defined.
- Document architecture decisions and material trade-offs in concise Architecture Decision Records (ADRs).

14.3 Technology Decisions Requiring Approval

Decision	Options / proposal	Evaluation criteria
Cloud provider	AWS, Airtel Cloud or another approved provider	India region/data residency, service availability, security, support, cost, backups, network and team capability
Compute	Managed application platform, VM/container deployment or equivalent	Operational burden, scale, patching, deployment and cost
Database hosting	Managed PostgreSQL preferred for reduced operational burden, or self-managed where justified	Availability, backups, restore, patching, HA, monitoring and cost
Authentication	Approved JWT/session design or enterprise identity provider	Threat model, SSO needs, token lifecycle, MFA and supportability
File storage	Private object storage or approved secure file service	Encryption, access control, retention, signed URL policy, malware scanning and cost
Email delivery	Approved SMTP/API provider if notifications require email	Deliverability, domain setup, cost, logs and retry policy
CI/CD	Approved repository pipeline and deployment mechanism	Review gates, secret handling, rollback, environment promotion and auditability

The PRD does not commit the organization to a cloud provider or cost before a technical/commercial comparison. A deployment decision should include an estimated monthly operating cost, one-time setup effort, backup/restore design and ownership of patching and incidents.

15. Deployment, Environments and Operations

15.1 Environments

Environment	Purpose	Data / access rules
Development	Daily engineering and integration work	Synthetic data; developer access; no production secrets
Test / QA	Automated and manual validation	Synthetic or anonymized data; repeatable test fixtures
Staging / UAT	Business acceptance and release rehearsal	Production-like configuration; controlled access; approved test data
Production	Live employee and business operations	Restricted access, approved release process, monitoring, backups and incident ownership

15.2 CI/CD and Release Controls

- Build and test on each pull request or approved change.
- Run unit/integration tests, static analysis and dependency checks.
- Require code review and protected-branch rules for production-bound changes.
- Deploy to staging before production; obtain business acceptance where required.
- Apply database migrations in a controlled, reversible or forward-fix-aware manner.
- Use health checks and post-deployment smoke tests.
- Maintain a rollback/roll-forward plan and identify the release owner.
- Keep environment-specific configuration and secrets outside source code.

15.3 Backup and Recovery

The team must define backup frequency, retention, encryption, access control, cross-zone/region strategy where appropriate, and restore ownership. Backup completion alone is not sufficient: restore tests must verify data integrity and application usability.

Item	Requirement
RPO	Maximum acceptable data loss window — TBD with management and business owners.
RTO	Maximum acceptable service recovery time — TBD with management and business owners.
Backup policy	Frequency, retention, encryption, storage location and deletion controls — to be approved.
Restore testing	Documented restore drill before go-live and at a recurring approved cadence.
Incident response	Named escalation contacts, severity definitions, communications and recovery runbook.

15.4 Monitoring and Support

- Monitor application health, API error rates, latency, database health, disk/storage capacity and certificate/secret expiry where applicable.
- Use structured logs with correlation IDs and defined retention.
- Alert on service unavailability, repeated failures, backup failures and security-relevant events.
- Define severity levels, response ownership, escalation paths and support hours.
- Document deployment, rollback, database restore, credential rotation and incident triage runbooks.

16. Testing Strategy and Acceptance Criteria

16.1 Test Layers

Test layer	Coverage
Unit tests	Business rules, validations, date calculations, leave balance logic, payroll formulas and permission utilities.
Integration tests	Spring services, repository/database interactions, transaction boundaries, migrations and API behaviour.
API contract tests	Request validation, response schemas, status codes, pagination, error formats and authorization.
End-to-end tests	Employee onboarding, attendance regularization, leave approval, payroll preparation and payslip access.
Security tests	Authentication, role/record access, direct API abuse, injection, session/token handling and sensitive export restrictions.
Performance tests	Representative list, search, report, dashboard and payroll workloads using agreed data volumes.
Recovery tests	Backup restoration, deployment rollback/roll-forward and recovery runbooks.
UAT	HR, Finance, managers and selected employees validate approved workflows against acceptance criteria.

16.2 Release Acceptance Gates

- All P0 requirements have linked, passing test cases or an explicitly approved exception.
- No unresolved critical security defect; high-risk exceptions require formal documented acceptance.
- Payroll test scenarios reconcile exactly according to approved rounding and calculation rules.
- Role and record-level access tests demonstrate that users cannot access unauthorized records through direct API calls.
- Database migrations have been tested against a representative environment and data volume.
- Backup restore and incident/recovery procedures have been rehearsed.
- Monitoring, alerting, logs and operational ownership are in place.
- HR and Finance sign off on relevant business rules; management approves go-live readiness.
- Known issues, support ownership, rollback plan and release notes are documented.

16.3 Essential Test Scenarios

- Duplicate employee ID; invalid department reference; employee deactivation with historical records.

- Manager attempts to view a non-team employee; employee attempts to retrieve another employee's payslip.
- Two approvers act on the same request concurrently; only one valid terminal transition is recorded.
- Leave requests overlap; balance is insufficient; approver is inactive; request is cancelled after approval.
- Attendance crosses midnight; punch is missing; duplicate import row exists; regularization is approved.
- Salary structure changes mid-period; employee joins or leaves mid-period; a payroll input is missing.
- Payroll calculation is retried; run is finalized; post-finalization correction is attempted.
- Database connection fails during a transaction; transaction rolls back without partial balance or payroll updates.
- Backup restore is performed; restored application can authenticate and retrieve reconciled test data.
- API receives malformed input, oversized upload or repeated authentication attempts.

17. Delivery Roadmap and Governance

Phase	Primary deliverables	Exit / approval gate
0 — Discovery and decisions	Approved goals, multi-tenant SaaS architecture decision, user roles, policy workshops, scope, business rules, deployment comparison and risk register	Management, HR and Finance scope & architecture sign-off
1 — Foundation	Repository, multi-tenant architecture baseline, tenant isolation context, database migrations, API standards, authentication/authorization foundation, feature entitlement model, policy versioning core, CI/CD and environments	Security, multi-tenancy and architecture review
2 — Organization and employee core	Organization setup, employee lifecycle, user mapping, onboarding and audit events	Employee workflow UAT
3 — Attendance and leave	Shifts, attendance records/import foundation, regularization, leave policies/balances/approvals	HR workflow UAT and reconciliation
4 — Payroll and payslips	Salary structures, approved calculations, payroll runs, exceptions, finalization and payslips	Finance rule approval and payroll reconciliation
5 — Reporting and hardening	Reports, notifications, performance/security testing, monitoring, backup and recovery drills	QA and production readiness review
6 — Pilot and go-live	Controlled pilot, issue resolution, training, migration, support handover and launch	Formal go-live approval
Later releases	Approved integrations, advanced reporting and additional HR capabilities	Separate prioritization and change approval

17.1 Governance and Ownership

Role	Accountability
Management Sponsor	Business priority, budget, scope trade-offs and final go-live approval.
Product Owner / IT Lead	Requirements baseline, backlog prioritization, cross-team coordination, risks and delivery reporting.
HR Process Owner	Employee lifecycle, attendance, leave and HR policy decisions.
Finance / Payroll Owner	Salary components, payroll formulas, reconciliation and statutory rule validation.
Technical Lead / Backend	Architecture, APIs, database integrity, security implementation and technical documentation.
Frontend Engineering	Accessible UI, API integration, validation, responsive behaviour and frontend tests.
QA / Test Owner	Test strategy, traceability, regression coverage, defect triage and release evidence.
Cloud / Operations Owner	Hosting, deployments, monitoring, backups, access and incident response.

17.2 Change Control

Every material change to approved scope should include the business rationale, affected requirement IDs, priority, effort and schedule impact, dependencies, risks, acceptance criteria and approving owner. Unapproved additions should not be treated as committed delivery.

18. Risks, Dependencies and Mitigations

Risk / dependency	Impact	Mitigation / owner
HR policies not finalized	Rework or incorrect leave/attendance behaviour	HR process owner approves rules before implementation.
Payroll requirements incomplete	Incorrect pay outcomes and compliance exposure	Finance validates rules, examples, effective dates and test cases.
Cloud provider undecided	Cost or architecture rework	Complete a weighted comparison before production environment commitment.
Scope creep	Delivery delay and reduced quality	P0/P1/P2 backlog and formal change control.
Sensitive data mishandled	Privacy/security incident	Data classification, least privilege, encryption, audit and restricted exports.
Insufficient test coverage	Production defects or inconsistent balances	Requirement traceability, automated tests, UAT and release gates.

Risk / dependency	Impact	Mitigation / owner
Data migration quality	Duplicate or incomplete employee records	Data profiling, mapping, validation, dry runs and reconciliation if migration is in scope.
Integration provider instability	Workflow delays or duplicate events	Timeouts, retries, idempotency, monitoring and manual reconciliation path.
No recovery rehearsal	Long outage or data loss	Approved RPO/RTO, scheduled restore drills and runbooks.
Unclear operational ownership	Unresolved incidents and maintenance gaps	Assign service owner, support hours, escalation and release responsibility before go-live.

19. Assumptions, Open Decisions and Change Control

19.1 Current Planning Assumptions

- The product is intended for production use and must use a backend and persistent relational database.
- React/Vite, Spring Boot/Java 21 and PostgreSQL are the proposed baseline, subject to technical approval.
- The first release focuses on core HRMS workflows rather than every possible HR software category.
- Payroll functionality must be driven by approved company policy and validated applicable requirements.
- Cloud hosting will be selected after security, regional availability, operations and cost review.
- The application must support role-based and record-level access from the first production release.
- Development and test environments should use synthetic or anonymized data.

19.2 Open Decisions Register

ID	Decision required	Owner	Status
OD-01	Single-company deployment or multi-tenant product architecture — Resolved: Multi-tenant SaaS with shared database, row-level tenant isolation, and tenant-scoped configuration.	Management / IT	Approved / Closed
OD-02	AWS vs Airtel Cloud vs other provider; region and hosting model	IT / Management	Open
OD-03	Authentication approach, SSO/MFA requirements and account recovery	IT / Security	Open
OD-04	Approved employee fields, mandatory documents and onboarding workflow	HR	Open
OD-05	Attendance rules, shift patterns, grace periods and import format	HR	Open

ID	Decision required	Owner	Status
OD-06	Leave categories, accrual, carry-forward, expiry and approval matrix	HR	Open
OD-07	Salary components, payroll periods, proration and rounding	Finance	Open
OD-08	Applicable statutory rules and external expert validation	Finance / HR	Open
OD-09	Required reports and sensitive-field visibility	Management / HR / Finance	Open
OD-10	Data retention, backups, RPO/RTO and incident response	IT / Management	Open
OD-11	Initial data migration scope and source systems	HR / IT	Open
OD-12	Go-live date, pilot group, support hours and service targets	Management / Product Owner	Open

19.3 Decision Management

Open decisions must be assigned an owner and target date during planning. Decisions affecting payroll, privacy, security, data architecture or cloud cost must be resolved before the related implementation is considered production-ready.

20. Acceptance, Handover and Sign-Off

20.1 Product Acceptance

Product acceptance requires agreement that the approved P0 scope is implemented, tested, documented and demonstrated; role-based access and data integrity are verified; HR and Finance approve their relevant business rules; and operational readiness gates are met.

20.2 Handover Package

- Approved PRD and change log.
- Architecture diagram and key architecture decisions.
- Database ERD and migration history.
- API specification and integration notes.
- Role-permission matrix and security configuration guidance.
- Test plan, test results, defects and UAT sign-off.

- Deployment and rollback runbooks.
- Backup/restore and incident-response runbooks.
- Administrator and user guides.
- Known limitations, support contacts and future backlog.

20.3 Sign-Off

Stakeholder	Approval responsibility	Name / signature / date
Management Sponsor	Business goals, scope, priorities and go-live readiness	Pending
HR Process Owner	Employee, attendance and leave requirements	Pending
Finance / Payroll Owner	Payroll rules, calculations and reconciliation	Pending
IT Lead / Product Owner	Requirements baseline, delivery plan and coordination	Pending
Technical / Operations Owner	Architecture, security, deployment and recovery readiness	Pending

Approval statement: Sign-off confirms the agreed scope and requirements baseline. It does not waive applicable security, privacy, financial-control or statutory obligations.

Appendix A — Requirement ID Catalogue

Requirement IDs are stable references for backlog items, implementation tickets, test cases, defect reports and change requests. Do not renumber IDs after they are referenced by downstream artifacts.

Prefix	Domain
FR-IAM	Identity, authentication and authorization
FR-ORG / FR-EMP	Organization and employee lifecycle
FR-ATT	Attendance and shifts
FR-LEV	Leave management
FR-PAY	Payroll and payslips
FR-ESS	Employee self-service
FR-REP	Reports and dashboards
FR-NOT	Notifications
FR-AUD / FR-ADM	Audit, administration and support
BR-*	Business rules
OD-*	Open decisions

Prefix	Domain
NFR-*	Non-functional requirements; detailed numeric targets to be approved

Appendix B — Suggested API Resource Catalogue

This is a resource-level planning aid, not a final API contract. Detailed schemas, pagination conventions, error codes, authorization requirements and examples should be defined in OpenAPI before implementation.

- Authentication: /api/v1/auth
- Users and roles: /api/v1/users, /api/v1/roles, /api/v1/permissions
- Organization: /api/v1/organizations, /api/v1/branches, /api/v1/departments, /api/v1/designations
- Employees: /api/v1/employees, /api/v1/employees/{id}/documents
- Attendance: /api/v1/attendance, /api/v1/attendance/imports, /api/v1/attendance-requests
- Shifts and calendars: /api/v1/shifts, /api/v1/shift-assignments, /api/v1/holidays
- Leave: /api/v1/leave-types, /api/v1/leave-policies, /api/v1/leave-balances, /api/v1/leave-requests
- Payroll: /api/v1/salary-components, /api/v1/salary-structures, /api/v1/payroll-periods, /api/v1/payroll-runs
- Payslips: /api/v1/payslips
- Reports: /api/v1/reports/employees, /attendance, /leave, /payroll
- Notifications: /api/v1/notifications
- Audit: /api/v1/audit-events
- Operational health: /actuator/health with production-safe exposure configuration

Appendix C — Production Readiness Checklist

- [ ] Management, HR and Finance have approved the release scope and applicable rules.
- [ ] All P0 requirements are traceable to implementation and passing tests.
- [ ] Role and record-level authorization tests pass for all sensitive modules.
- [ ] Payroll scenarios reconcile with Finance-approved examples.
- [ ] Database migrations, indexes and constraints are reviewed and tested.
- [ ] Secrets are externalized; no production credentials are present in source control.
- [ ] TLS, encryption, access controls and log redaction are verified.
- [ ] Dependency/security scans are reviewed and release-blocking findings resolved.
- [ ] Backups are configured and a restore test is evidenced.
- [ ] Monitoring, alerting, health checks and incident ownership are active.
- [ ] Deployment and rollback/roll-forward procedures are tested.
- [ ] UAT, user training, support documentation and known-issue register are complete.
- [ ] Data migration is reconciled if migration is in scope.
- [ ] Final go-live approval is recorded.

Appendix D — Vendor Feature Scope & Release Classification Matrix

Purpose: compare the supplied HRMS vendor feature catalogue with the current production PRD baseline. This matrix prevents vendor capability from being mistaken for an approved delivery commitment.

Classification rules

- Included in Phase 1 — explicitly part of the proposed core production baseline. Detailed business rules and acceptance criteria still require the relevant owner's approval.
- Planned for Phase 2/3 — not committed to the initial release; subject to prioritization, dependencies, estimates and formal approval.
- Out of Scope / Not Committed — not part of the current roadmap unless a change request is approved. Important boundary: a module name appearing in the PRD does not mean every advanced feature in that module is included. In particular, basic payroll is not equivalent to validated statutory tax computation or automated government filing.

ESS Web Portal & Mobile App

Feature / Capability	Scope Status	Scope Note
ESS web portal for employee self-service	Included in Phase 1	Core employee self-service is in the baseline; exact screens and permissions follow approved ESS requirements.
Dedicated native mobile app / mobile-specific workflows	Planned for Phase 2/3	A dedicated app is not a Phase 1 deliverable; mobile-responsive web is the initial approach unless approved otherwise.

Core HR

Feature / Capability	Scope Status	Scope Note
Multiple entities / SBUs with separate configuration	Included in Phase 1	Tenant context, multi-entity / SBU hierarchy, and tenant-scoped configuration baselined in Phase 1 foundation per Architecture Review.
Multiple branches	Included in Phase 1	Multi-branch management with location-specific calendars and policy assignments baselined in Phase 1 foundation.
Department management	Included in Phase 1	Part of the proposed core production baseline; detailed rules and acceptance criteria still require owner sign-off.
Role and access management	Included in Phase 1	Part of the proposed core production baseline; detailed rules and acceptance criteria still require owner sign-off.
Custom field management	Planned for Phase 2/3	Advanced configurability is not explicitly committed in the current Phase 1 requirements.

Feature / Capability	Scope Status	Scope Note
HCM setup / organization configuration	Included in Phase 1	Part of the proposed core production baseline; detailed rules and acceptance criteria still require owner sign-off.
Footer and signature setup	Planned for Phase 2/3	Advanced configurability is not explicitly committed in the current Phase 1 requirements.

Employee Management

Feature / Capability	Scope Status	Scope Note
Employee onboarding workflow	Included in Phase 1	Part of the proposed core production baseline; detailed rules and acceptance criteria still require owner sign-off.
Bulk import employee details	Planned for Phase 2/3	Not fully specified in the Phase 1 acceptance criteria; schedule as a later enhancement unless HR approves it for the initial release.
Organization chart visualization	Planned for Phase 2/3	Not fully specified in the Phase 1 acceptance criteria; schedule as a later enhancement unless HR approves it for the initial release.
Employee hierarchy chart	Included in Phase 1	Part of the proposed core production baseline; detailed rules and acceptance criteria still require owner sign-off.
Employee profile management	Included in Phase 1	Part of the proposed core production baseline; detailed rules and acceptance criteria still require owner sign-off.
Employee self-onboarding	Planned for Phase 2/3	Not fully specified in the Phase 1 acceptance criteria; schedule as a later enhancement unless HR approves it for the initial release.
Employee profile update approval workflow	Planned for Phase 2/3	Not fully specified in the Phase 1 acceptance criteria; schedule as a later enhancement unless HR approves it for the initial release.
Employee job description management	Planned for Phase 2/3	Not fully specified in the Phase 1 acceptance criteria; schedule as a later enhancement unless HR approves it for the initial release.
Employee documentation management	Planned for Phase 2/3	Not fully specified in the Phase 1 acceptance criteria; schedule as a later enhancement unless HR approves it for the initial release.

Attendance Management

Feature / Capability	Scope Status	Scope Note
Attendance policy setup	Included in Phase 1	Part of the proposed core production baseline; detailed rules and acceptance criteria still require owner sign-off.
Overtime policy setup	Planned for Phase 2/3	Advanced attendance rule or device/mobile capability not fully baselined; confirm policy, device/provider and acceptance tests before scheduling.

Feature / Capability	Scope Status	Scope Note
Attendance email notifications	Planned for Phase 2/3	Advanced attendance rule or device/mobile capability not fully baselined; confirm policy, device/provider and acceptance tests before scheduling.
Attendance import / bulk upload	Planned for Phase 2/3	Advanced attendance rule or device/mobile capability not fully baselined; confirm policy, device/provider and acceptance tests before scheduling.
Rotational shift management	Planned for Phase 2/3	Advanced attendance rule or device/mobile capability not fully baselined; confirm policy, device/provider and acceptance tests before scheduling.
Flexi shifts	Planned for Phase 2/3	Advanced attendance rule or device/mobile capability not fully baselined; confirm policy, device/provider and acceptance tests before scheduling.
Night shift management	Planned for Phase 2/3	Advanced attendance rule or device/mobile capability not fully baselined; confirm policy, device/provider and acceptance tests before scheduling.
Roster / work schedule management	Planned for Phase 2/3	Advanced attendance rule or device/mobile capability not fully baselined; confirm policy, device/provider and acceptance tests before scheduling.
Geo-fencing	Planned for Phase 2/3	Advanced attendance rule or device/mobile capability not fully baselined; confirm policy, device/provider and acceptance tests before scheduling.
Geo-location validation	Planned for Phase 2/3	Advanced attendance rule or device/mobile capability not fully baselined; confirm policy, device/provider and acceptance tests before scheduling.
Attendance correction and regularization	Included in Phase 1	Part of the proposed core production baseline; detailed rules and acceptance criteria still require owner sign-off.
Employee movement register	Planned for Phase 2/3	Advanced attendance rule or device/mobile capability not fully baselined; confirm policy, device/provider and acceptance tests before scheduling.
Biometric integration	Planned for Phase 2/3	Advanced attendance rule or device/mobile capability not fully baselined; confirm policy, device/provider and acceptance tests before scheduling.
Automatic present marking	Planned for Phase 2/3	Advanced attendance rule or device/mobile capability not fully baselined; confirm policy, device/provider and acceptance tests before scheduling.
Multiple punch-in handling	Planned for Phase 2/3	Advanced attendance rule or device/mobile capability not fully baselined; confirm policy, device/provider and acceptance tests before scheduling.

Feature / Capability	Scope Status	Scope Note
Web app punch-in	Included in Phase 1	Part of the proposed core production baseline; detailed rules and acceptance criteria still require owner sign-off.
Mobile app punch-in	Planned for Phase 2/3	Advanced attendance rule or device/mobile capability not fully baselined; confirm policy, device/provider and acceptance tests before scheduling.
Selfie-based punch-in	Planned for Phase 2/3	Advanced attendance rule or device/mobile capability not fully baselined; confirm policy, device/provider and acceptance tests before scheduling.

Leave Management

Feature / Capability	Scope Status	Scope Note
Leave policy setup	Included in Phase 1	Part of the proposed core production baseline; detailed rules and acceptance criteria still require owner sign-off.
Leave clubbing rules	Planned for Phase 2/3	Specific rule configuration is not fully specified in the Phase 1 baseline; requires HR policy confirmation and test cases.
Leave entitlement configuration	Included in Phase 1	Part of the proposed core production baseline; detailed rules and acceptance criteria still require owner sign-off.
Holiday master	Included in Phase 1	Part of the proposed core production baseline; detailed rules and acceptance criteria still require owner sign-off.
Holiday allocation list	Included in Phase 1	Part of the proposed core production baseline; detailed rules and acceptance criteria still require owner sign-off.
Leave application import	Planned for Phase 2/3	Specific rule configuration is not fully specified in the Phase 1 baseline; requires HR policy confirmation and test cases.
Leave carry-forward rules	Planned for Phase 2/3	Specific rule configuration is not fully specified in the Phase 1 baseline; requires HR policy confirmation and test cases.
Leave encashment rules	Planned for Phase 2/3	Specific rule configuration is not fully specified in the Phase 1 baseline; requires HR policy confirmation and test cases.
Leave sandwich rules	Planned for Phase 2/3	Specific rule configuration is not fully specified in the Phase 1 baseline; requires HR policy confirmation and test cases.
Notice-period leave rules	Planned for Phase 2/3	Specific rule configuration is not fully specified in the Phase 1 baseline; requires HR policy confirmation and test cases.
Compensatory-off request	Included in Phase 1	Part of the proposed core production baseline; detailed rules and acceptance criteria still require owner sign-off.
Compensatory-off payment	Planned for Phase 2/3	Specific rule configuration is not fully specified in the Phase 1 baseline; requires HR policy confirmation and test cases.

Feature / Capability	Scope Status	Scope Note
Leave approval workflows	Included in Phase 1	Part of the proposed core production baseline; detailed rules and acceptance criteria still require owner sign-off.
Leave application restrictions	Planned for Phase 2/3	Specific rule configuration is not fully specified in the Phase 1 baseline; requires HR policy confirmation and test cases.

Payroll Management

Feature / Capability	Scope Status	Scope Note
Company / SBU-wise payroll setup	Planned for Phase 2/3	Advanced payroll or statutory functionality not fully specified in the initial baseline; Finance validation and approved calculation rules are required.
Payroll cycle setup	Included in Phase 1	Part of the proposed core production baseline; detailed rules and acceptance criteria still require owner sign-off.
Compliance branch setup	Planned for Phase 2/3	Advanced payroll or statutory functionality not fully specified in the initial baseline; Finance validation and approved calculation rules are required.
Pay elements setup	Included in Phase 1	Part of the proposed core production baseline; detailed rules and acceptance criteria still require owner sign-off.
Automatic location-wise payroll compliance	Planned for Phase 2/3	Advanced payroll or statutory functionality not fully specified in the initial baseline; Finance validation and approved calculation rules are required.
Salary structure templates	Included in Phase 1	Part of the proposed core production baseline; detailed rules and acceptance criteria still require owner sign-off.
Employee salary details	Included in Phase 1	Part of the proposed core production baseline; detailed rules and acceptance criteria still require owner sign-off.
Employee compliance setup	Planned for Phase 2/3	Advanced payroll or statutory functionality not fully specified in the initial baseline; Finance validation and approved calculation rules are required.
Payroll adjustments	Included in Phase 1	Part of the proposed core production baseline; detailed rules and acceptance criteria still require owner sign-off.
Incentives / retention bonus management	Planned for Phase 2/3	Advanced payroll or statutory functionality not fully specified in the initial baseline; Finance validation and approved calculation rules are required.
Salary increment arrears	Planned for Phase 2/3	Advanced payroll or statutory functionality not fully specified in the initial baseline; Finance validation and approved calculation rules are required.

Feature / Capability	Scope Status	Scope Note
One-click payroll processing	Included in Phase 1	Part of the proposed core production baseline; detailed rules and acceptance criteria still require owner sign-off.
Salary hold	Included in Phase 1	Part of the proposed core production baseline; detailed rules and acceptance criteria still require owner sign-off.
Import previous-month salary data	Planned for Phase 2/3	Advanced payroll or statutory functionality not fully specified in the initial baseline; Finance validation and approved calculation rules are required.
Form 24Q	Planned for Phase 2/3	Advanced payroll or statutory functionality not fully specified in the initial baseline; Finance validation and approved calculation rules are required.

TDS & Income Tax

Feature / Capability	Scope Status	Scope Note
TDS cycle setup	Planned for Phase 2/3	Requires a separately approved tax specification, current rules, Finance validation, effective-date handling and statutory test cases.
Income-tax regime selection	Planned for Phase 2/3	Requires a separately approved tax specification, current rules, Finance validation, effective-date handling and statutory test cases.
Automatic TDS calculation	Planned for Phase 2/3	Requires a separately approved tax specification, current rules, Finance validation, effective-date handling and statutory test cases.
Investment declaration and proof management	Planned for Phase 2/3	Requires a separately approved tax specification, current rules, Finance validation, effective-date handling and statutory test cases.
Tax projection and simulation	Planned for Phase 2/3	Requires a separately approved tax specification, current rules, Finance validation, effective-date handling and statutory test cases.
Form 16 generation and consolidated pay sheet	Planned for Phase 2/3	Requires a separately approved tax specification, current rules, Finance validation, effective-date handling and statutory test cases.
F&F TDS computation	Planned for Phase 2/3	Requires a separately approved tax specification, current rules, Finance validation, effective-date handling and statutory test cases.
Perquisite management	Planned for Phase 2/3	Requires a separately approved tax specification, current rules, Finance validation, effective-date handling and statutory test cases.

Loans & Advances

Feature / Capability	Scope Status	Scope Note
Salary advance / loan policy setup	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.

Feature / Capability	Scope Status	Scope Note
Salary advance / loan request workflow	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Interest and charge configuration	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Loan integration with payroll	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Disbursement and EMI configuration	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Loan balance and statement tracking	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.

Expense Management

Feature / Capability	Scope Status	Scope Note
Expense policy setup	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Expense claim submission	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Expense claim approval workflow	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Expense reports	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.

Travel Management

Feature / Capability	Scope Status	Scope Note
Travel policy setup	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Travel request and approval workflow	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Travel advance management	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Travel booking	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Travel expense payment submission	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.

Confirmation & Probation

Feature / Capability	Scope Status	Scope Note
Confirmation / probation process setup	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Automated probation tracking	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Configurable confirmation workflow	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Confirmation evaluation form	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Confirmation decision and actions	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Confirmation notifications and reminders	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Document and letter automation for confirmation	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Contractor renewal tracking	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.

Promotion & Transfer Management

Feature / Capability	Scope Status	Scope Note
Centralized promotion and transfer workflow	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Promotion management	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Transfer management	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.

Feature / Capability	Scope Status	Scope Note
Promotion / transfer communication and letters	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Promotion / transfer audit and history	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Bulk promotion / transfer option	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.

OKR & Goals

Feature / Capability	Scope Status	Scope Note
Define OKR cycle	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Assign OKR cycle to employees	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
OKR templates	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
OKR review frequency setup	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
OKR / goal setup and approval workflow	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
OKR / goal review and approval workflow	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
OKR notifications and reminders	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Track OKR progress	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
OKR dashboards	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
OKR reports	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.

Separation Management

Feature / Capability	Scope Status	Scope Note
Separation process setup	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Separation workflow setup	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Exit interview and feedback	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Exit checklist automation	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Full and final settlement workflow	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Separation analytics and reporting	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.

Document Management

Feature / Capability	Scope Status	Scope Note
Payslip generation	Included in Phase 1	Part of the proposed core production baseline; detailed rules and acceptance criteria still require owner sign-off.
HR letter template setup	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
HR policy document upload	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Digital employee letter repository	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Employee documentation audit	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.

Employee Helpdesk

Feature / Capability	Scope Status	Scope Note
Employee helpdesk setup	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Configurable query categories and SLAs	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.

Feature / Capability	Scope Status	Scope Note
Employee query / ticket management	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
SLA tracking and escalation	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Helpdesk analytics and reports	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.

Task Management

Feature / Capability	Scope Status	Scope Note
Customized task template setup	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Centralized task assignment and tracking	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Progress tracking and status management	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Daily task management	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Task reports and analytics	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.

Timesheet Management

Feature / Capability	Scope Status	Scope Note
Customized timesheet template setup	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Project-wise time tracking	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Automated timesheet workflow	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Billability and utilization tracking	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.

Feature / Capability	Scope Status	Scope Note
Timesheet integration with payroll and attendance	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Timesheet reports and analytics	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.

Email Notification

Feature / Capability	Scope Status	Scope Note
SMTP setup	Included in Phase 1	Part of the proposed core production baseline; detailed rules and acceptance criteria still require owner sign-off.
Email notification setup	Included in Phase 1	Part of the proposed core production baseline; detailed rules and acceptance criteria still require owner sign-off.
Timed reminders	Included in Phase 1	Part of the proposed core production baseline; detailed rules and acceptance criteria still require owner sign-off.
Mail event setup	Included in Phase 1	Part of the proposed core production baseline; detailed rules and acceptance criteria still require owner sign-off.
Compose and send email from the application	Planned for Phase 2/3	A configurable compose-email feature is not explicitly included in the initial baseline; core system notifications remain in scope.

Offer & Pre-Onboarding

Feature / Capability	Scope Status	Scope Note
Offer letter template setup	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Digital offer and candidate management	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Pre-joining data collection	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Joining formalities automation	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Joining and compliance form automation	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.

Asset Management

Feature / Capability	Scope Status	Scope Note
Centralized asset repository	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Asset request and return workflow	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Employee-wise asset allocation	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Asset lifecycle tracking	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Inventory and stock management	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Asset condition and maintenance tracking	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.
Asset renewal and insurance due dates	Planned for Phase 2/3	Not part of the initial release baseline. Requires prioritization, detailed requirements, dependencies and approval before commitment.

Learning Management

Feature / Capability	Scope Status	Scope Note
Centralized learning hub	Out of Scope / Not Committed	Learning management is outside the current core HRMS roadmap and requires a separate business case and approval.
Course creation and management	Out of Scope / Not Committed	Learning management is outside the current core HRMS roadmap and requires a separate business case and approval.
Learning calendar and notifications	Out of Scope / Not Committed	Learning management is outside the current core HRMS roadmap and requires a separate business case and approval.
Learning paths and skill development	Out of Scope / Not Committed	Learning management is outside the current core HRMS roadmap and requires a separate business case and approval.
Assessments and certifications	Out of Scope / Not Committed	Learning management is outside the current core HRMS roadmap and requires a separate business case and approval.

Master Reports — Attendance & Leave Reports

Feature / Capability	Scope Status	Scope Note
Monthly attendance report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Daily present / absent report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Attendance muster roll report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Attendance muster roll with leave-code details	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Attendance penalty report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Attendance summary report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Employee attendance report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Missing punch report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Geo-fencing reports	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Leave report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Compensatory-off report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Applied leave report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.

Feature / Capability	Scope Status	Scope Note
Leave balance report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Monthly leave report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Employee late-coming / early-going report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Work-schedule assignment details	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.

Master Reports — Statutory & Enrollment Reports

Feature / Capability	Scope Status	Scope Note
LWF statement	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
LWF Form A-1	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
PF statement	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
PF register	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
PF return	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
ESIC statement	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
ESIC return	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.

Feature / Capability	Scope Status	Scope Note
ESIC register	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
PT statement	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
PT Form 5	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
New-joiner EPF enrollment report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
New-joiner ESIC enrollment report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
NPS statement	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Gratuity projection	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Form D — Equal Remuneration Act	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Form D — Equal Remuneration Act gender-wise	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.

Master Reports — Salary, Payroll & Arrears Reports

Feature / Capability	Scope Status	Scope Note
Employee-type-wise salary sheet	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Employee-type-wise salary register with overtime and penalty	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.

Feature / Capability	Scope Status	Scope Note
Employee-type-wise salary register	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Employee-type-wise CTC register	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Employee-type-wise salary comparison sheet	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Department-wise salary summary report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
JV salary report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
JV reimbursement report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
SBU and branch-wise summary report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
SBU-wise salary register report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Branch-wise salary register report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Branch-wise salary summary report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
SBU-wise salary summary report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Payslips	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Employee salary structure	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.

Feature / Capability	Scope Status	Scope Note
Monthly salary summary report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
New-joiner monthly salary report, SBU-wise	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Left-employee monthly salary report, SBU-wise	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Employee-type-wise CTC salary register	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Employee-type-wise salary register with arrears	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Arrears summary report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Monthly arrears disbursement report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Net payable summary report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.

Master Reports — Banking & Remittance Reports

Feature / Capability	Scope Status	Scope Note
HDFC bank report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Pay-type-wise HDFC bank report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Bank-wise bank report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.

Feature / Capability	Scope Status	Scope Note
Bank report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Bank balance credit report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Element-wise bank report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Bank remittance report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Bank statement report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Department and sub-department-wise bank report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.

Master Reports — Tax Reports

Feature / Capability	Scope Status	Scope Note
Projected TDS report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Month-wise TDS liability report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Monthly TDS statement	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
TDS declaration report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Monthly TDS report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.

Feature / Capability	Scope Status	Scope Note
HRA declaration report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Housing-loan declaration report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.

Master Reports — Employee, Organization & Administration Reports

Feature / Capability	Scope Status	Scope Note
Role assignment list	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Organization-wise headcount details	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Employee probation details	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Employee special-moment report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Pending reporting-person assignment	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Manpower detail report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Organization education detail	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Organization gender detail	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Employee joining and left-service report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.

Feature / Capability	Scope Status	Scope Note
Employee growth history	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Leave encashment report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Organization age-wise detail	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Organization experience-wise detail	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Employee history	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Employee login report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Credential and login report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Ticket report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
PMS status report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Employee defaulter report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Employee-wise weekly timesheet report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.
Monthly timesheet report	Planned for Phase 2/3	The PRD currently commits to basic operational reporting, not this full named report catalogue. Each report needs fields, filters, permissions, export format and reconciliation tests approved.

Reporting Baseline

Feature / Capability	Scope Status	Scope Note
Basic employee / attendance / leave / payroll operational reports required for Phase 1	Included in Phase 1	Baseline reports are limited to the approved core workflows; do not assume every vendor-listed report is included.

Cross-Module Capabilities

Feature / Capability	Scope Status	Scope Note
Configurable approval workflows across all modules	Included in Phase 1 (Core Engine) / Phase 2 (Visual Designer)	Configurable multi-step approval workflows (manager/HR chains, delegation, escalation) included in Phase 1 foundation; visual drag-and-drop workflow designer planned for Phase 2.
Complete business-process history for every module	Planned for Phase 2/3	General audit logs are in baseline; complete module-specific business history is a later enhancement.
Full audit history for document and promotion / transfer events	Planned for Phase 2/3	Specialized audit views are not fully specified in the initial baseline.
Advanced dashboards and workforce analytics	Planned for Phase 2/3	Basic operational visibility is in scope; advanced analytics and configurable dashboards are later-release candidates.
Direct statutory filing / submission to government portals	Out of Scope / Not Committed	Automatic legal/tax filing is explicitly excluded unless validated rules, authorized integrations and formal approval are provided.
Bank payment execution / direct salary disbursement	Out of Scope / Not Committed	Direct bank payment execution is not part of the current initial-release commitment.

Management decision required

Approve the Phase 1 baseline, confirm whether multi-entity/branch support and advanced employee workflows are required at launch, and assign owners for Phase 2/3 modules. TDS/statutory calculations must not be declared production-ready until Finance validates the applicable rules and test cases. Any change to these classifications must follow the PRD change-control process.